



AHMET EMIR KALAFAT

Electrical-Electronics & Computer Engineer • Industrial Automation (PLC / SCADA)

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PROFILE

Electrical & Electronics Engineer (double major in Computer Engineering, Erasmus+ semester in Electrical and Automation Engineering) with hands-on **PLC programming in CODESYS (IEC 61131-3)**, STM32 embedded C and Raspberry Pi machine vision. Adds **3 years of professional software experience**, two years of it in an English-speaking international team at a German company, focused on system integration and Python tooling. Seeking an automation engineering role (PLC / SCADA / HMI, commissioning) where strong software skills strengthen control systems and industrial data integration.

TECHNICAL SKILLS

PLC & Automation	CODESYS – IEC 61131-3 (Ladder, FBD, IL, Structured Text)
Electrical Design	EPLAN, AutoCAD Electrical
Simulation & PCB	MATLAB, Proteus, LTspice, Multisim • KiCad
Embedded Systems	STM32 (C), Raspberry Pi 4, sensor integration, Bluetooth communication, motor control
Programming	C, C#, Java, Python (OpenCV, pandas, pydantic, Tkinter)
Software & Tools	Git / GitHub, AWS, React, React Native, Flutter, Firebase
Languages	Turkish (Native) • English (C1 – Advanced)

EDUCATION

B.Sc. Electrical and Electronics Engineering (30% English-medium • GPA 3.00 / 4.00) 2020 – 2024
Fatih Sultan Mehmet Vakif University – Istanbul, Türkiye

Erasmus+ Exchange – Electrical and Automation Engineering (taught in English) Jan 2023 – Jun 2023
Klaipėda State University of Applied Sciences (Klaipėdos valstybinė kolegija) – Klaipėda, Lithuania

B.Sc. Computer Engineering (Double Major) 2021 – 2025
Fatih Sultan Mehmet Vakif University – Istanbul, Türkiye

ENGINEERING PROJECTS

Barcode Reading System with Microprocessor – Graduation Project Raspberry Pi 4 • OpenCV

- Designed and built a camera-based barcode reader on a **Raspberry Pi 4** with an **OV5647 5 MP camera module**.
- Developed a Python / **OpenCV** image-processing pipeline (thresholding, erosion, dilation) for noise-free images.
- Added an operator-controlled image capture function to reduce errors caused by incorrect exposure; decoded pixel data in line with the **EAN-13** standard – suitable for retail, logistics and production tracking.

Bluetooth-Controlled Smart Vehicle STM32 • C • Flutter

- Programmed an **STM32 microcontroller in C** for real-time motor control, 360° rotation and sensor data processing.
- Implemented a **collision-avoidance safety interlock**: front/rear sensors detect obstacles and override operator input.
- Built a Flutter control app, integrated with the hardware over Bluetooth (movement, headlight control).
- Worked closely with teammates to bring hardware, firmware and the mobile app together into one working system – a project that strengthened my **teamwork and cross-disciplinary communication** skills.

PROFESSIONAL EXPERIENCE

Talsen Team GmbH – Full-Stack Software Engineer Freilassing, Germany • Jul 2024 – Jul 2026

- Core developer of the **ARA project management software** across frontend (React), backend / API and the Python **ara-cli** tool.
- Integrated frontend, API and CLI into one system in an international team using English as the daily working language, following a strict agile process.
- Built LLM-agent-based code generation infrastructure integrated into ARA to automate development workflows.

SameUp – Mobile App Developer (Intern → Part-time) Istanbul • Jul 2023 – Jul 2024

- Led the UI/UX overhaul of the **Kidokit** app, implementing a new design language across 20+ core screens (React Native).
- Co-developed the **Investag** investment tracking app from scratch, building 14+ key features and reusable components.

ADDITIONAL INFORMATION

Activities: New Generation R&D – Board Member; My University Lecture Notes Mobile Application – Moderator

Side Project: Tarif Atölyesi – recipe-sharing mobile app (Flutter, Firebase), published on Google Play and the App Store